Full Stack Assignment – 18

Assignment ID: FSD-18

Title: Task Assignment Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Task Assignment Management System** that allows managers to assign tasks and team members to track them.

The system must enforce scheduling rules and ensure **data persistence using a database**.

2️⃣ User Roles

MANAGER

- Register and log in to the system
- Create and assign tasks
- View all tasks and assignments

MEMBER

- Register and log in to the system
- View assigned tasks
- Update task progress (optional/basic)

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Task Management

- MANAGER can create tasks with start date and deadline
- Task data must be stored in the database
- Tasks must be assigned to members persistently

Task Assignment

- MANAGER assigns tasks to MEMBERS
- MEMBERS can view their assigned tasks
- Assignment data must reflect real-time database state

4️⃣ Business Rule (Critical)

⚠️ **A task deadline must not be earlier than its assignment date.**

Tasks violating this rule must be rejected at creation time.

5️⃣ Technical Constraints (Mandatory)

- All task and assignment data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for MANAGER and MEMBER
- Task creation and assignment view for MANAGER
- Task listing view for MEMBER

UI design is not evaluated; correctness and data flow matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints

- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **date validation, role enforcement, and end-to-end execution**.